

lab02(SSU)

September 8, 2025

```
[1]: !apt-get install texlive texlive-xetex texlive-latex-extra pandoc
!pip install pypandoc
```

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

The following additional packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java
libcmark-gfm-extensions0.29.0.gfm.3 libcmark-gfm0.29.0.gfm.3
libcommons-logging-java libcommons-parent-java libfontbox-java libgs9
libgs9-common libidn12 libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java
libptexenc1 libruby3.0 libsynchronet2 libteckit0 libtexlua53 libtexluajit2
libwoff1 libzip-0-13 lmodern pandoc-data poppler-data preview-latex-style
rake ruby ruby-net-telnet ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0
rubygems-integration tlutils teckit tex-common tex-gyre texlive-base
texlive-binaries texlive-fonts-recommended texlive-latex-base
texlive-latex-recommended texlive-pictures texlive-plain-generic tipa
xfonts-encodings xfonts-utils

Suggested packages:

fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
texlive-luatex pandoc-citeproc context wkhtmltopdf librsvg2-bin groff ghc
nodejs php python libjs-mathjax libjs-katex citation-style-language-styles
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-pstricks
dot2tex prerex texlive-pictures-doc vprerex default-jre-headless tipa-doc

The following NEW packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java
libcmark-gfm-extensions0.29.0.gfm.3 libcmark-gfm0.29.0.gfm.3
libcommons-logging-java libcommons-parent-java libfontbox-java libgs9

```

libgs9-common libidn12 libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java
libptexenc1 libruby3.0 libsynchronet2 libteckit0 libtexlua53 libtexluajit2
libwoff1 libzip-0-13 lmodern pandoc pandoc-data poppler-data
preview-latex-style rake ruby ruby-net-telnet ruby-rubygems ruby-webrick
ruby-xmlrpc ruby3.0 rubygems-integration tltutils teckit tex-common tex-gyre
texlive texlive-base texlive-binaries texlive-fonts-recommended
texlive-latex-base texlive-latex-extra texlive-latex-recommended
texlive-pictures texlive-plain-generic texlive-xetex tipa xfonts-encodings
xfonts-utils
0 upgraded, 58 newly installed, 0 to remove and 35 not upgraded.
Need to get 202 MB of archives.
After this operation, 728 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-droid-fallback all
1:6.0.1r16-1.1build1 [1,805 kB]
Get:2 http://security.ubuntu.com/ubuntu jammy-security/main amd64 ruby3.0 amd64
3.0.2-7ubuntu2.11 [50.1 kB]
Get:3 http://security.ubuntu.com/ubuntu jammy-security/main amd64 libruby3.0
amd64 3.0.2-7ubuntu2.11 [5,114 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-lato all 2.0-2.1
[2,696 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/main amd64 poppler-data all
0.4.11-1 [2,171 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-common all 6.17
[33.7 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-urw-base35 all
20200910-1 [6,367 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9-common
all 9.55.0~dfsg1-0ubuntu5.12 [753 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libidn12 amd64
1.38-4ubuntu1 [60.0 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy/main amd64 libijs-0.35 amd64
0.35-15build2 [16.5 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy/main amd64 libjbig2dec0 amd64
0.19-3build2 [64.7 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9 amd64
9.55.0~dfsg1-0ubuntu5.12 [5,031 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libkpathsea6
amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy/main amd64 libwoff1 amd64
1.0.2-1build4 [45.2 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy/universe amd64 dvisvgm amd64
2.13.1-1 [1,221 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-lmodern all
2.004.5-6.1 [4,532 kB]
Get:17 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-noto-mono all
20201225-1build1 [397 kB]
Get:18 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-texgyre all
20180621-3.1 [10.2 MB]

```

Get:19 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libapache-pom-java
all 18-1 [4,720 B]
Get:20 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcmark-
gfm0.29.0.gfm.3 amd64 0.29.0.gfm.3-3 [115 kB]
Get:21 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcmark-gfm-
extensions0.29.0.gfm.3 amd64 0.29.0.gfm.3-3 [25.1 kB]
Get:22 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-parent-
java all 43-1 [10.8 kB]
Get:23 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-logging-
java all 1.2-2 [60.3 kB]
Get:24 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libptexenc1
amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
Get:25 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration
all 1.18 [5,336 B]
Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-rubygems all
3.3.5-2 [228 kB]
Get:27 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby amd64 1:3.0~exp1
[5,100 B]
Get:28 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7
kB]
Get:29 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all
0.1.1-2 [12.6 kB]
Get:30 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick
all 1.7.0-3ubuntu0.2 [52.5 kB]
Get:31 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmlrpc all
0.3.2-1ubuntu0.1 [24.9 kB]
Get:32 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libsynctex2
amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
Get:33 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libteckit0 amd64
2.5.11+ds1-1 [421 kB]
Get:34 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexlua53
amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
Get:35 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexluajit2
amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
Get:36 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libzzip-0-13 amd64
0.13.72+dfsg.1-1.1 [27.0 kB]
Get:37 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-encodings all
1:1.0.5-0ubuntu2 [578 kB]
Get:38 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-utils amd64
1:7.7+6build2 [94.6 kB]
Get:39 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 lmodern all
2.004.5-6.1 [9,471 kB]
Get:40 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 pandoc-data all
2.9.2.1-3ubuntu2 [81.8 kB]
Get:41 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 pandoc amd64
2.9.2.1-3ubuntu2 [20.3 MB]
Get:42 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 preview-latex-style
all 12.2-1ubuntu1 [185 kB]

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Get:43 http://archive.ubuntu.com/ubuntu jammy/main amd64 t1utils amd64
1.41-4build2 [61.3 kB]
Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 teckit amd64
2.5.11+ds1-1 [699 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-gyre all
20180621-3.1 [6,209 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 texlive-
binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive all
2021.20220204-1 [14.3 kB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:54 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:55 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:56 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:57 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:58 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 202 MB in 7s (28.3 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 126374 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...

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Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.12_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.12) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.12_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.12) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcmark-gfm0.29.0.gfm.3:amd64.
Preparing to unpack .../17-libcmark-gfm0.29.0.gfm.3_0.29.0.gfm.3-3_amd64.deb ...
Unpacking libcmark-gfm0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Selecting previously unselected package libcmark-gfm-
extensions0.29.0.gfm.3:amd64.
Preparing to unpack .../18-libcmark-gfm-
extensions0.29.0.gfm.3_0.29.0.gfm.3-3_amd64.deb ...

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Unpacking libcmark-gfm-extensions0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../19-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../20-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../21-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../22-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../23-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../24-ruby-rubygems_3.3.5-2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../25-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../26-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../27-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../28-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../29-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../30-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../31-libsyntax2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../32-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../33-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...

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Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
.../34-libtexluajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libzzip-0-13:amd64.
Preparing to unpack .../35-libzzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../36-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../37-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../38-lmodern_2.004.5-6.1_all.deb ...
Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package pandoc-data.
Preparing to unpack .../39-pandoc-data_2.9.2.1-3ubuntu2_all.deb ...
Unpacking pandoc-data (2.9.2.1-3ubuntu2) ...
Selecting previously unselected package pandoc.
Preparing to unpack .../40-pandoc_2.9.2.1-3ubuntu2_amd64.deb ...
Unpacking pandoc (2.9.2.1-3ubuntu2) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../41-preview-latex-style_12.2-1ubuntu1_all.deb ...
Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package tlutils.
Preparing to unpack .../42-tlutils_1.41-4build2_amd64.deb ...
Unpacking tlutils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../43-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../44-tex-gyre_20180621-3.1_all.deb ...
Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../45-texlive-
binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../46-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack .../47-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../48-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...

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Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../49-texlive-latex-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive.
Preparing to unpack .../50-texlive_2021.20220204-1_all.deb ...
Unpacking texlive (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../51-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.
Preparing to unpack .../52-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...
Selecting previously unselected package texlive-pictures.
Preparing to unpack .../53-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../54-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../55-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../56-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../57-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libjbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...

```



```

Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webbrick (1.7.0-3ubuntu0.2) ...
Setting up libcmark-gfm0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up libcmark-gfm-extensions0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up pandoc-data (2.9.2.1-3ubuntu2) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.12) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.12) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up pandoc (2.9.2.1-3ubuntu2) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...

```

```

Setting up tipa (2:1.3-21) ...
Setting up texlive (2021.20220204-1) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.11) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmulubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

Processing triggers for tex-common (6.17) ...
Running updmap-sys. This may take some time... done.
Running mktexlsr /var/lib/texmf ... done.

```

```
Building format(s) --all.
```

```
    This may take some time... done.
```

```
Collecting pydoc
```

```
  Downloading pydoc-1.15-py3-none-any.whl.metadata (16 kB)
```

```
Downloading pydoc-1.15-py3-none-any.whl (21 kB)
```

```
Installing collected packages: pydoc
```

```
Successfully installed pydoc-1.15
```

```
[2]: # Connect Google Drive to Colab so you can access your files
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

1 Part 1. Numbers

Quantitative information arises everywhere in data science. In addition to representing commands to print out lines, expressions can represent numbers and methods of combining numbers. The expression `3.2500` evaluates to the number `3.25`. (Run the cell and see.)

```
[3]: 3.2500
```

```
[3]: 3.25
```

Notice that we didn't have to `print`. When you run a notebook cell, if the last line has a value, then Jupyter helpfully prints out that value for you. However, it won't print out prior lines automatically.

Above, you should see that `4` is the value of the last expression, `2` is printed, but `3` is lost forever because it was neither printed nor last.

You don't want to print everything all the time anyway. But if you feel sorry for `3`, change the cell above to print it.

```
[9]: print(2)
3
4
```

```
2
```

```
[9]: 4
```

1.1 1.1. Arithmetic

The line in the next cell subtracts. Its value is what you'd expect. Run it.

```
[10]: 3.25 - 1.5
```

```
[10]: 1.75
```

Many basic arithmetic operations are built into Python. The textbook section on [Expressions](#) describes all the arithmetic operators used in the course. The common operator that differs from typical math notation is `**`, which raises one number to the power of the other. So, `2**3` stands for 2^3 and evaluates to 8.

The order of operations is the same as what you learned in elementary school, and Python also has parentheses. For example, compare the outputs of the cells below. The second cell uses parentheses for a happy new year!

```
[11]: 9+6*5-6*3**2*2**3/4*7
```

```
[11]: -717.0
```

```
[12]: 9+(6*5-(6*3))**2*((2**3)/4*7)
```

```
[12]: 2025.0
```

In standard math notation, the first expression is

$$9 + 6 \times 5 - 6 \times 3^2 \times \frac{2^3}{4} \times 7,$$

while the second expression is

$$9 + (6 \times 5 - (6 \times 3))^2 \times \left(\frac{(2^3)}{4} \times 7 \right).$$

Question 1.1.1. Write a Python expression in this next cell that's equal to $5 \times \left(3\frac{10}{11}\right) - 50\frac{1}{3} + 2^{0.5 \times 22} - \frac{7}{33} + 8$. That's five times three and ten elevenths, minus fifty and a third, plus two to the power of half twenty-two, minus seven thirty-thirds plus seven.

Replace the ellipses (...) with your expression. Try to use parentheses only when necessary.

Note: By “ $3\frac{10}{11}$ ”, we mean $3 + \frac{10}{11}$, not $3 \times \frac{10}{11}$.

Hint: The correct output should start with a familiar number.

```
[13]: 5 * (3+10/11) - 50+1/3 + 2**(0.5*22) - 7/33 + 8
```

```
[13]: 2025.6666666666667
```

2 2. Names

In natural language, we have terminology that lets us quickly reference very complicated concepts. We don't say, “That's a large mammal with brown fur and sharp teeth!” Instead, we just say, “Bear!”

In Python, we do this with *assignment statements*. An assignment statement has a name on the left side of an = sign and an expression to be evaluated on the right.

```
[15]: ten = 3 * 2 + 4
```

When you run that cell, Python first computes the value of the expression on the right-hand side, $3 * 2 + 4$, which is the number 10. Then it assigns that value to the name `ten`. At that point, the code in the cell is done running.

After you run that cell, the value 10 is bound to the name `ten`:

```
[16]: ten
```

```
[16]: 10
```

The statement `ten = 3 * 2 + 4` is not asserting that `ten` is already equal to $3 * 2 + 4$, as we might expect by analogy with math notation. Rather, that line of code changes what `ten` means; it now refers to the value 10, whereas before it meant nothing at all.

If the designers of Python had been ruthlessly pedantic, they might have made us write

define the name `ten` to hereafter have the value of $3 * 2 + 4$

instead. You will probably appreciate the brevity of “=”! But keep in mind that this is the real meaning.

Question 2.1. Run the following cell which uses a variable name `eleven` that hasn’t been assigned to anything. You’ll see an error!

```
[17]: eleven + 8
```

```
-----  
NameError                                Traceback (most recent call last)  
/tmp/ipython-input-1097191323.py in <cell line: 0>()  
----> 1 eleven + 8  
  
NameError: name 'eleven' is not defined
```

A common pattern in Jupyter notebooks is to assign a value to a name and then immediately evaluate the name in the last line in the cell so that the value is displayed as output.

```
[18]: close_to_pi = 355/113  
      close_to_pi
```

```
[18]: 3.1415929203539825
```

Another common pattern is that a series of lines in a single cell will build up a complex computation in stages, naming the intermediate results.

```
[19]: semimonthly_salary = 843 + 3/4 # 843.75
      monthly_salary = 2 * semimonthly_salary
      number_of_months_in_a_year = 12
      yearly_salary = number_of_months_in_a_year * monthly_salary
      yearly_salary
```

```
[19]: 20250.0
```

Names in Python can have letters (upper- and lower-case letters are both okay and count as different letters), underscores, and numbers. The first character can't be a number (otherwise a name might look like a number). And names can't contain spaces, since spaces are used to separate pieces of code from each other.

Other than those rules, what you name something doesn't matter *to Python*. For example, this cell does the same thing as the above cell, except everything has a different name:

```
[20]: a = 842.5
      b = 2 * a
      c = 12
      d = c * b
      d
```

```
[20]: 20220.0
```

However, names are very important for making your code *readable* to yourself and others. The cell above is shorter, but it's totally useless without an explanation of what it does.

2.1 2.1. Checking Your Code

Now that you know how to name things, you can start using the built-in *tests* to check whether your work is correct. Sometimes, there are multiple tests for a single question, and passing all of them is required to receive credit for the question. Please don't change the contents of the test cells.

Go ahead and attempt Question 2.1.2. Running the cell directly after it will test whether you have assigned `seconds_in_a_decade` correctly in Question 2.1.2. If you haven't, this test will tell you the correct answer. Resist the urge to just copy it, and instead try to adjust your expression. (Sometimes the tests will give hints about what went wrong...)

Question 2.1.2. Assign the name `seconds_in_a_decade` to the number of seconds between midnight January 1, 2010 and midnight January 1, 2020. Note that there are two leap years in this span of a decade. A non-leap year has 365 days and a leap year has 366 days.

Hint: If you're stuck, the next section shows you how to get hints.

```
[24]: # Change the next line
      # so that it computes the number of seconds in a decade
      # and assigns that number the name, seconds_in_a_decade.
```

```
seconds_in_a_decade = 24 * 60 * 60 * (365 * 10 + 2)

# We've put this line in this cell
# so that it will print the value you've given to seconds_in_a_decade when you
↳ run it.
# You don't need to change this.
seconds_in_a_decade
```

[24]: 315532800

3 2.2. Comments

You may have noticed these lines in the cell in which you answered Question 2.2:

```
# Change the next line
# so that it computes the number of seconds in a decade
# and assigns that number the name, seconds_in_a_decade.
```

This is called a *comment*. It doesn't make anything happen in Python; Python ignores anything on a line after a `#`. Instead, it's there to communicate something about the code to you, the human reader. Comments are extremely useful.

Source: [XKCD](#)

3.1 2.3. Application: A Physics Experiment

On the Apollo 15 mission to the Moon, astronaut David Scott famously replicated Galileo's physics experiment in which he showed that gravity accelerates objects of different mass at the same rate. Because there is no air resistance for a falling object on the surface of the Moon, even two objects with very different masses and densities should fall at the same rate. David Scott compared a feather and a hammer.

You can run the following cell to watch a video of the experiment.

```
[25]: from IPython.display import YouTubeVideo
# The original URL is:
# https://www.youtube.com/watch?v=U7db6ZeLR5s

YouTubeVideo("U7db6ZeLR5s")
```

[25]:



Here's the transcript of the video:

167:22:06 Scott: Well, in my left hand, I have a feather; in my right hand, a hammer. And I guess one of the reasons we got here today was because of a gentleman named Galileo, a long time ago, who made a rather significant discovery about falling objects in gravity fields. And we thought where would be a better place to confirm his findings than on the Moon. And so we thought we'd try it here for you. The feather happens to be, appropriately, a falcon feather for our Falcon. And I'll drop the two of them here and, hopefully, they'll hit the ground at the same time.

167:22:43 Scott: How about that!

167:22:45 Allen: How about that! (Applause in Houston)

167:22:46 Scott: Which proves that Mr. Galileo was correct in his findings.

Newton's Law. Using this footage, we can also attempt to confirm another famous bit of physics: Newton's law of universal gravitation. Newton's laws predict that any object dropped near the surface of the Moon should fall

$$\frac{1}{2}G\frac{M}{R^2}t^2 \text{ meters}$$

after t seconds, where G is a universal constant, M is the moon's mass in kilograms, and R is the moon's radius in meters. So if we know G , M , and R , then Newton's laws let us predict how far

an object will fall over any amount of time.

To verify the accuracy of this law, we will calculate the difference between the predicted distance the hammer drops and the actual distance. (If they are different, it might be because Newton's laws are wrong, or because our measurements are imprecise, or because there are other factors affecting the hammer for which we haven't accounted.)

Someone studied the video and estimated that the hammer was dropped 113 cm from the surface. Counting frames in the video, the hammer falls for 1.2 seconds (36 frames).

Question 2.3.1. Complete the code in the next cell to fill in the data from the experiment.

Hint: No computation required; just fill in data from the paragraph above.

```
[28]: # t, the duration of the fall in the experiment, in seconds.
# Fill this in.
time = 1.2

# The estimated distance the hammer actually fell, in meters.
# Fill this in.
estimated_distance_m = 1.13
```

Question 2.3.2. Now, complete the code in the next cell to compute the difference between the predicted and estimated distances (in meters) that the hammer fell in this experiment.

This just means translating the formula above ($\frac{1}{2}G\frac{M}{R^2}t^2$) into Python code. You'll have to replace each variable in the math formula with the name we gave that number in Python code.

Hint: Try to use variables you've already defined in question 2.3.1

```
[29]: # First, we've written down the values of the 3 universal constants
# that show up in Newton's formula.

# G, the universal constant measuring the strength of gravity.
gravity_constant = 6.674 * 10**-11

# M, the moon's mass, in kilograms.
moon_mass_kg = 7.34767309 * 10**22

# R, the radius of the moon, in meters.
moon_radius_m = 1.737 * 10**6

# The distance the hammer should have fallen
# over the duration of the fall, in meters,
# according to Newton's law of gravity.
# The text above describes the formula
# for this distance given by Newton's law.
# **YOU FILL THIS PART IN.**
predicted_distance_m = 1/2 * gravity_constant * moon_mass_kg / moon_radius_m**2
    ↪ * time**2
```

```
# Here we've computed the difference
# between the predicted fall distance and the distance we actually measured.
# If you've filled in the above code, this should just work.
difference = predicted_distance_m - estimated_distance_m
difference
```

[29]: 0.040223694659304865

3.2 3. Calling Functions

The most common way to combine or manipulate values in Python is by calling functions. Python comes with many built-in functions that perform common operations.

For example, the `abs` function takes a single number as its argument and returns the absolute value of that number. Run the next two cells and see if you understand the output.

[30]: `abs(5)`

[30]: 5

[31]: `abs(-5)`

[31]: 5

3.3 3.1. Application: Computing Walking Distances

Chunhua is on the corner of 7th Avenue and 42nd Street in Midtown Manhattan, and she wants to know far she'd have to walk to get to Gramercy School on the corner of 10th Avenue and 34th Street.

She can't cut across blocks diagonally, since there are buildings in the way. She has to walk along the sidewalks. Using the map below, she sees she'd have to walk 3 avenues (long blocks) and 8 streets (short blocks). In terms of the given numbers, she computed 3 as the difference between 7 and 10, *in absolute value*, and 8 similarly.

Chunhua also knows that blocks in Manhattan are all about 80m by 274m (avenues are farther apart than streets). So in total, she'd have to walk $(80 \times |42 - 34| + 274 \times |7 - 10|)$ meters to get to the park.

Question 3.1.1. Fill in the line `num_avenues_away = ...` in the next cell so that the cell calculates the distance Chunhua must walk and gives it the name `manhattan_distance`. Everything else has been filled in for you. **Use the `abs` function.** Also, be sure to run the test cell afterward to test your code.

[32]:

```
# Here's the number of streets away:
num_streets_away = abs(42-34)

# Compute the number of avenues away in a similar way:
```

```

num_avenues_away = abs(7-10)

street_length_m = 80
avenue_length_m = 274

# Now we compute the total distance Chunhua must walk.
manhattan_distance = street_length_m*num_streets_away +
    ↪avenue_length_m*num_avenues_away

# We've included this line so that you see the distance you've computed
# when you run this cell.
# You don't need to change it, but you can if you want.
manhattan_distance

```

[32]: 1462

Multiple arguments Some functions take multiple arguments, separated by commas. For example, the built-in `max` function returns the maximum argument passed to it.

```
[34]: max(2, -3, 4, -5)
```

[34]: 4

4 4. Understanding Nested Expressions

Function calls and arithmetic expressions can themselves contain expressions. You saw an example in the last question:

```
abs(42-34)
```

has 2 number expressions in a subtraction expression in a function call expression. And you probably wrote something like `abs(7-10)` to compute `num_avenues_away`.

Nested expressions can turn into complicated-looking code. However, the way in which complicated expressions break down is very regular.

Suppose we are interested in lengths of cats that are very unusual. We'll say that a length is unusual to the extent that it's far away on the number line from the average cat length. An estimate of the average cat length (averaging, we hope, over all cats on Earth today) is **18.2** inches.

So if Ravioli is 21.7 inches long, then her length is $|21.7 - 18.2|$, or 3.5, inches away from the average. Here's a picture of that:

The source for average cat length is [Wikipedia](#). The listed lengths for cats are not real and may not be plausible (but the names are of real cats!)

And here's how we'd write that expression in one line of Python code:

```
[35]: abs(21.7 - 18.2)
```

[35]: 3.5

What's going on here? `abs` takes just one argument, so the stuff inside the parentheses is all part of that *single argument*. Specifically, the argument is the value of the expression `21.7 - 18.2`. The value of that expression is 3.5. That value is the argument to `abs`. The absolute value of that is 3.5, so 3.5 is the value of the full expression `abs(21.7 - 18.2)`.

Picture simplifying the expression in several steps:

1. `abs(21.7 - 18.2)`
2. `abs(3.5)`
3. 3.5

In fact, that's basically what Python does to compute the value of the expression.

Question 4.1. Say that Genghis's length is 16.7 inches. In the next cell, use `abs` to compute the absolute value of the difference between Genghis's length and the average cat length. Give that value the name `genghis_distance_from_average_in`.

```
[36]: # Replace the ... with an expression
      # to compute the absolute value
      # of the difference between Genghis's length (16.7 in) and the average cat_
      ↪length.
      genghis_distance_from_average_in = abs(18.2-16.7)

      # Again, we've written this here
      # so that the distance you compute will get printed
      # when you run this cell.
      genghis_distance_from_average_in
```

[36]: 1.5

4.1 4.1. More Nesting

Now say that we want to compute the more unusual of the two cat lengths. We'll use the function `max`, which (again) takes two numbers as arguments and returns the larger of the two arguments. Combining that with the `abs` function, we can compute the larger distance from average among the two length:

```
[37]: # Just read and run this cell.

      ravioli_length_in = 21.7
      genghis_length_in = 16.7
      average_cat_length = 18.2

      # The larger distance from the average cat length, among the two length:
      larger_distance_in = max(abs(ravioli_length_in - average_cat_length),
      ↪abs(genghis_length_in - average_cat_length))
```

```
# Print out our results in a nice readable format:
print("The larger distance from the average length among these two cats is",
      ↪larger_distance_in, "inches.")
```

The larger distance from the average length among these two cats is 3.5 inches.

The line where `larger_distance_in` is computed looks complicated, but we can break it down into simpler components just like we did before.

The basic recipe is to repeatedly simplify small parts of the expression: * **Basic expressions:** Start with expressions whose values we know, like names or numbers. - Examples: `genghis_length_in` or `16.7`. * **Find the next simplest group of expressions:** Look for basic expressions that are directly connected to each other. This can be by arithmetic or as arguments to a function call. - Example: `genghis_length_in - average_cat_length`. * **Evaluate that group:** Evaluate the arithmetic expression or function call. Use the value computed to replace the group of expressions. - Example: `genghis_length_in - average_cat_length` becomes `-1.3`. * **Repeat:** Continue this process, using the value of the previously-evaluated expression as a new basic expression. Stop when we've evaluated the entire expression. - Example: `abs(-1.3)` becomes `1.3`, and `max(3.5, 1.3)` becomes `3.5`.

You can run the next cell to see a slideshow of that process.

```
[38]: from IPython.display import IFrame
      IFrame('https://docs.google.com/presentation/d/
      ↪14EKhQjGe0bUpLwKGo6BrPCtjsORSkY8h4BzAqvXoLTM/embed?
      ↪start=false&loop=false&delayms=3000', 800, 600)
```

```
[38]: <IPython.lib.display.IFrame at 0x7dfe23258980>
```

Ok, your turn.

Question 4.1.1. Given the lengths of Yanay's cats Hummus, Gatkes, and Zeepty, write an expression that computes the smallest difference between any of the three lengths. Your expression shouldn't have any numbers in it, only function calls and the names `hummus`, `gatkes`, and `zeepty`. Give the value of your expression the name `min_length_difference`.

```
[39]: # The three cats' lengths, in inches:
      hummus = 24.5 # Hummus is 24.5 inches long
      gatkes = 19.7 # Gatkes is 19.7 inches long
      zeepty = 15.8 # Zeepty is 15.8 inches long

      # We'd like to look at all 3 pairs of lengths,
      # compute the absolute difference between each pair,
      # and then find the smallest of those 3 absolute differences.
      # This is left to you!
      # If you're stuck, try computing the value for each step of the process
      # (like the difference between Hummus's length and Gatkes's length)
      # on a separate line and giving it a name (like hummus_gatkes_length_diff)
```

```
min_length_difference =  
    ↪min(abs(hummus-gatkes),abs(hummus-zeepty),abs(gatkes-zeepty))  
min_length_difference
```

[39]: 3.8999999999999986

You're done with Lab 2!

4.2 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a pdf file for you to submit. **Please save before exporting!**

```
[ ]: # should change the directory and file name matching to yours  
!jupyter nbconvert '/content/drive/MyDrive/Colab Notebooks/lab02(SSU).ipynb'  
    ↪--to pdf
```